

Trotterized Quantum Simulation of Many-Body Hamiltonians: Theory, Error Structure, and Nuclear Physics Applications

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Digital quantum simulation relies on the efficient approximation of unitary time evolution generated by sums of noncommuting operators. Product-formula methods, commonly referred to as Lie–Trotter–Suzuki decompositions, provide a transparent and hardware-compatible route to implementing such dynamics on quantum processors. In this work we present a unified and rigorous analysis of Trotterized time evolution, emphasizing the operator-algebraic origin of simulation errors, their dependence on Hamiltonian structure, and their manifestation in quantum circuits and entanglement dynamics. We benchmark these features across three representative models: the transverse-field Ising model, the Lipkin–Meshkov–Glick model, and the nuclear pairing Hamiltonian. Using framework-independent circuit constructions and classical reference simulations based on NumPy and SciPy, we elucidate the interplay between locality, collectivity, and long-range correlations in determining Trotter convergence. Finally, we connect real-time Trotterization to variational quantum eigensolvers and unitary coupled cluster methods, establishing a common language for digital quantum simulation in condensed-matter, quantum chemistry, and nuclear physics.

I. INTRODUCTION

The controlled simulation of quantum many-body dynamics lies at the heart of quantum computing’s promise for physics, chemistry, and materials science. While analog quantum simulators provide direct access to specific Hamiltonians, digital quantum computers offer universality through gate-based implementations of unitary time evolution.

In practice, physically relevant Hamiltonians are expressed as sums of noncommuting terms, rendering exact implementation of the evolution operator e^{-iHt} impossible. Product-formula methods—originating in the work of Trotter and Suzuki—approximate this evolution by sequences of exponentials of simpler operators. These decompositions are not merely numerical conveniences: their error structure reflects the algebraic and physical properties of the Hamiltonian itself.

In this work, we develop a comprehensive analysis of Trotterized quantum simulation tailored to many-body Hamiltonians relevant to nuclear physics. We combine rigorous error bounds, explicit quantum-circuit constructions, and classical reference simulations to illuminate how interaction structure, connectivity, and entanglement generation govern digital simulation accuracy.

II. HAMILTONIAN DECOMPOSITION AND PAULI STRUCTURE

We consider Hamiltonians acting on n qubits that admit a Pauli expansion

$$H = \sum_{j=1}^N c_j P_j, \quad (1)$$

where each P_j is a tensor product of Pauli operators. Such decompositions arise naturally in lattice spin models, fermionic systems after Jordan–Wigner or Bravyi–Kitaev mappings, and quasi-spin representations of nuclear Hamiltonians.

If all P_j commute, the time-evolution operator factorizes exactly. In general, the nonvanishing commutators $[P_i, P_j]$ control the approximation error of product formulas.

III. LIE–TROTTER–SUZUKI PRODUCT FORMULAE

For bounded operators A and B , the first-order Lie–Trotter formula reads

$$e^{A+B} = \lim_{r \rightarrow \infty} \left(e^{A/r} e^{B/r} \right)^r. \quad (2)$$

At finite r , the leading error is proportional to the commutator $[A, B]$.

Higher-order Suzuki constructions systematically cancel lower-order commutator terms at the cost of increased circuit depth. The balance between algorithmic and hardware error is therefore central to practical quantum simulation.

IV. OPERATOR-ALGEBRAIC ORIGIN OF TROTTER ERROR

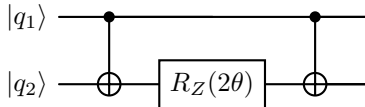
The Trotter error arises from the reordering of noncommuting operators in the Taylor expansion of e^{A+B} . Each swap introduces a commutator, yielding the bound

$$\epsilon \sim \frac{t^2}{r} \sum_{i < j} \|[H_i, H_j]\|. \quad (3)$$

This expression makes explicit that Trotter convergence depends not only on system size, but on the algebraic connectivity of the Hamiltonian.

V. QUANTUM CIRCUITS FOR PAULI EXPONENTIALS

Exponentials of Pauli strings can be implemented using single-qubit rotations and CNOT gates. For example,



implements $e^{-i\theta Z_1 Z_2}$. General Pauli strings follow by basis changes.

VI. MODEL HAMILTONIANS

A. Transverse-Field Ising Model

$$H_{\text{TFIM}} = J \sum_i Z_i Z_{i+1} - \Gamma \sum_i X_i. \quad (4)$$

B. Lipkin–Meshkov–Glick Model

$$H_{\text{LMG}} = \varepsilon J_z + V(J_x^2 - J_y^2), \quad J_\alpha = \frac{1}{2} \sum_i \sigma_i^\alpha. \quad (5)$$

C. Nuclear Pairing Hamiltonian

$$H_{\text{pair}} = 2 \sum_p \epsilon_p S_p^z - G \sum_{p,q} S_p^+ S_q^-. \quad (6)$$

Each model exhibits a distinct commutator structure, enabling systematic comparison of Trotter error scaling.

VII. COMPARATIVE ERROR SCALING

Local interactions in the TFIM yield linear error growth with system size. The all-to-all structure of the LMG model produces quadratic scaling. Pairing Hamiltonians interpolate between these regimes due to quasi-spin symmetries.

VIII. ENTANGLEMENT DYNAMICS

Entanglement entropy provides a sensitive diagnostic of simulation accuracy:

$$S_A = -\text{Tr}(\rho_A \log \rho_A). \quad (7)$$

Trotter errors typically manifest earlier in entanglement growth than in local observables, particularly for collective models.

IX. CONNECTION TO VQE AND UNITARY COUPLED CLUSTER

The unitary coupled cluster ansatz,

$$U_{\text{UCC}} = \exp \left(\sum_k \theta_k (\tau_k - \tau_k^\dagger) \right), \quad (8)$$

is implemented via Trotterized exponentials, making product-formula analysis directly relevant to variational quantum algorithms and imaginary-time methods.

X. REPRODUCIBILITY

All classical benchmarks can be reproduced using NumPy and SciPy. Circuit constructions are expressed independently of any quantum SDK, ensuring hardware- and framework-agnostic reproducibility.

XI. CONCLUSIONS AND OUTLOOK

Trotterized quantum simulation—when analyzed through the lens of operator algebra, entanglement, and model structure—reveals deep connections between digital quantum algorithms and traditional many-body physics. Nuclear models provide particularly rich testbeds for exploring these connections, positioning quantum computing as a natural extension of established theoretical frameworks.

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